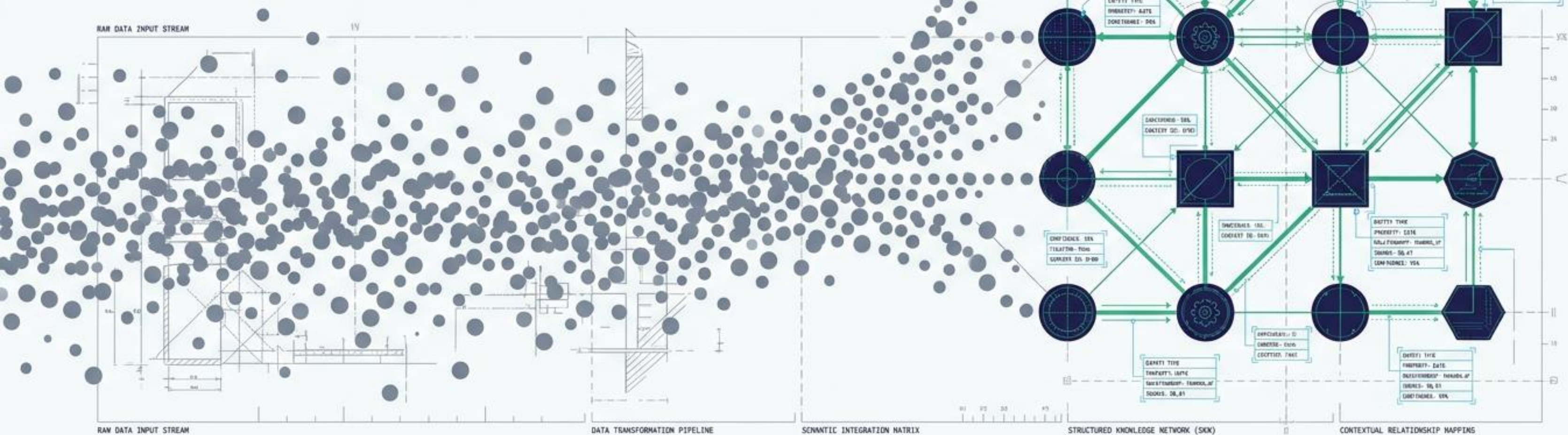
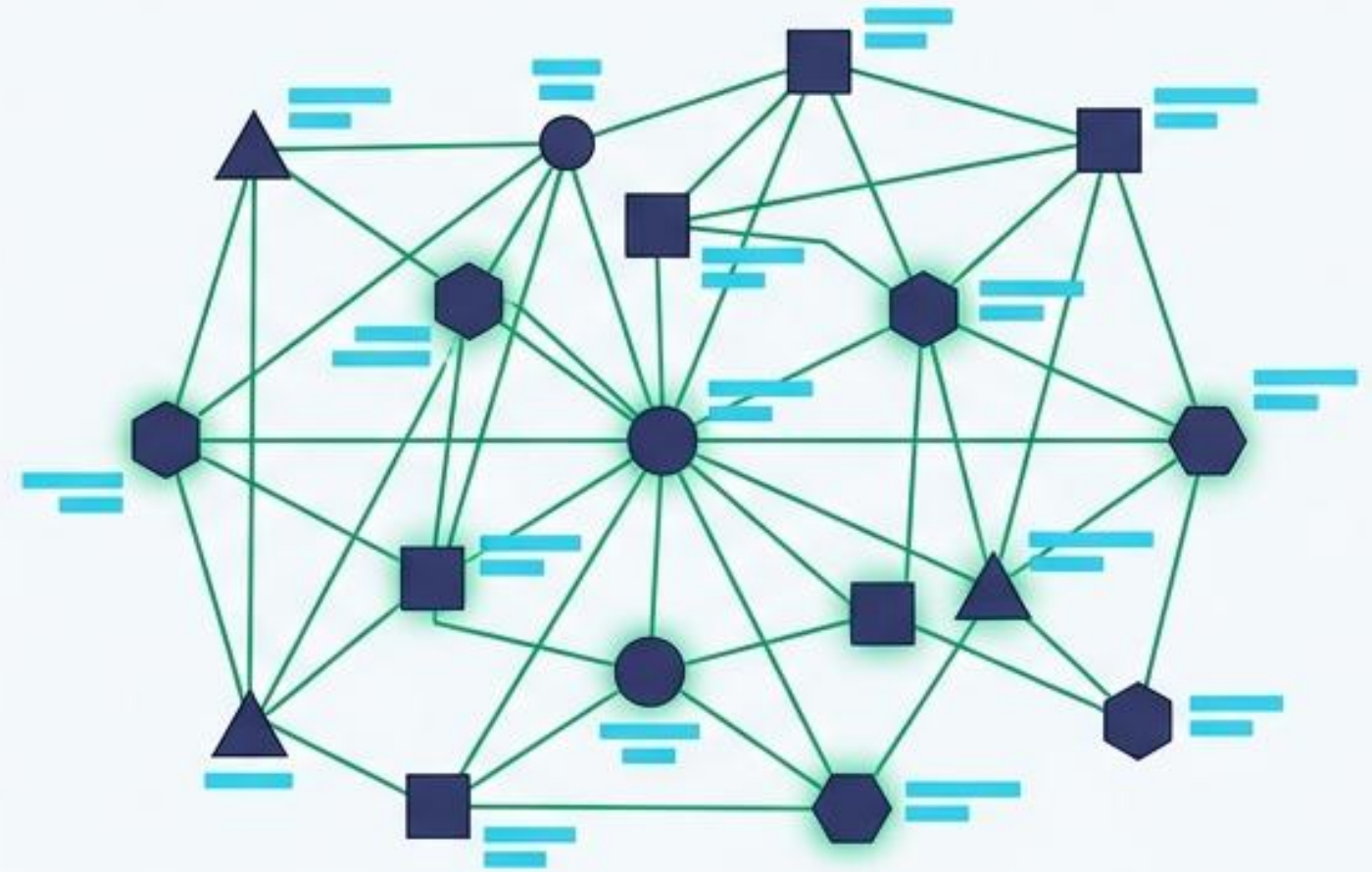


Introducing Knowledge Graphs

By Dr. Zbigniew Galar



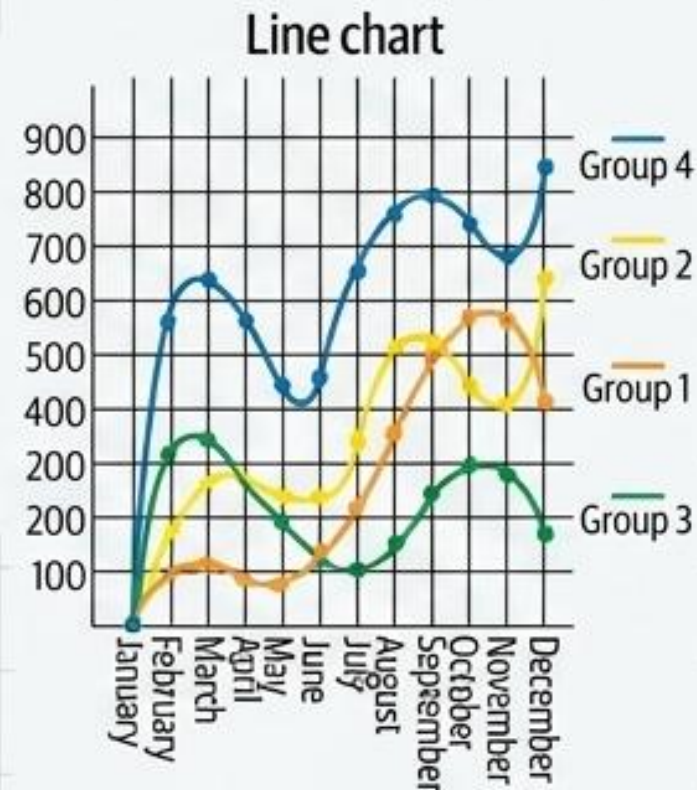
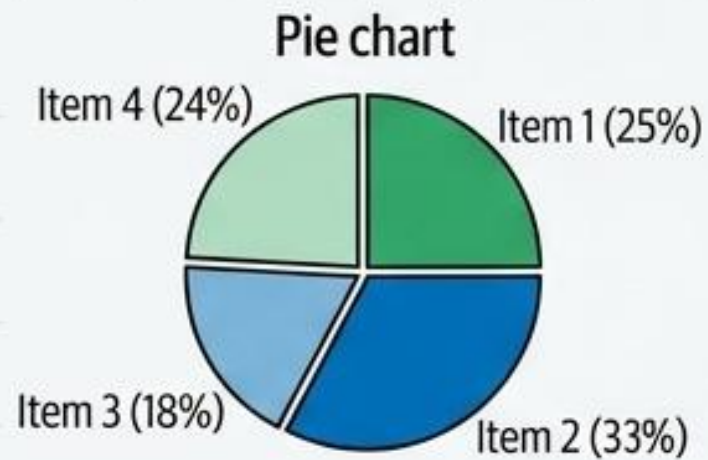
The Data Overwhelm and the Search for Context



Applying knowledge in context is the most powerful tool businesses have. A new category of technology based on graphs has emerged to capture this context and distill value from data.

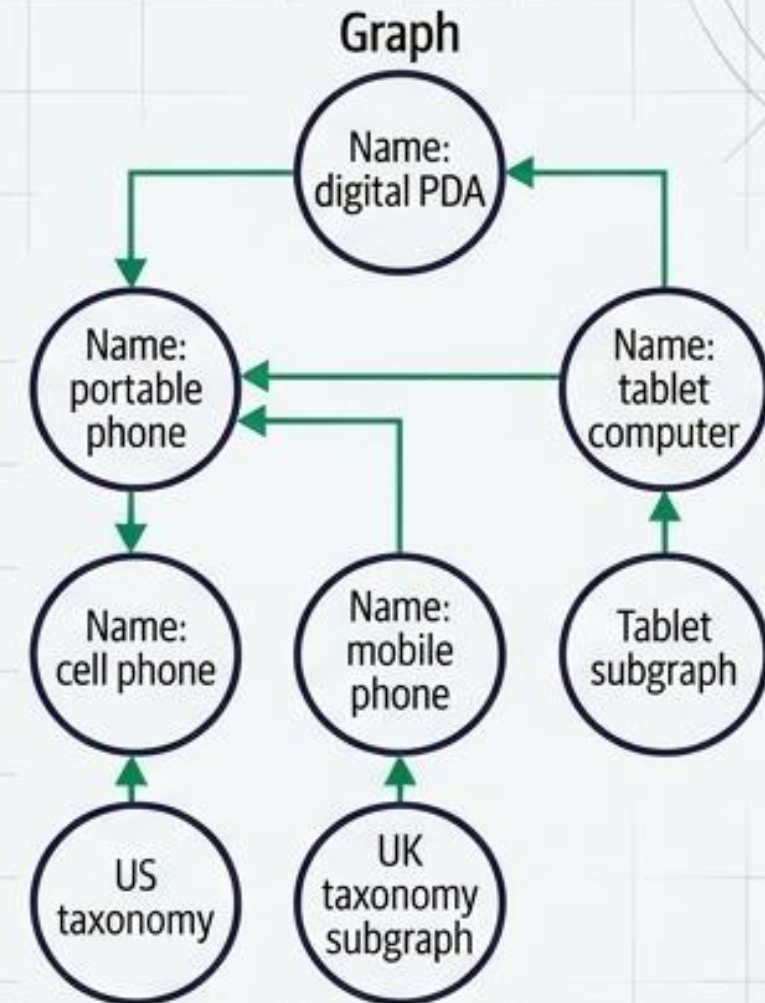
Distinguishing Charts from Graphs

Visualization Tools



Aggregating data, plotting functions, and displaying metrics (e.g., histograms, trends).

High-Fidelity Domain Models

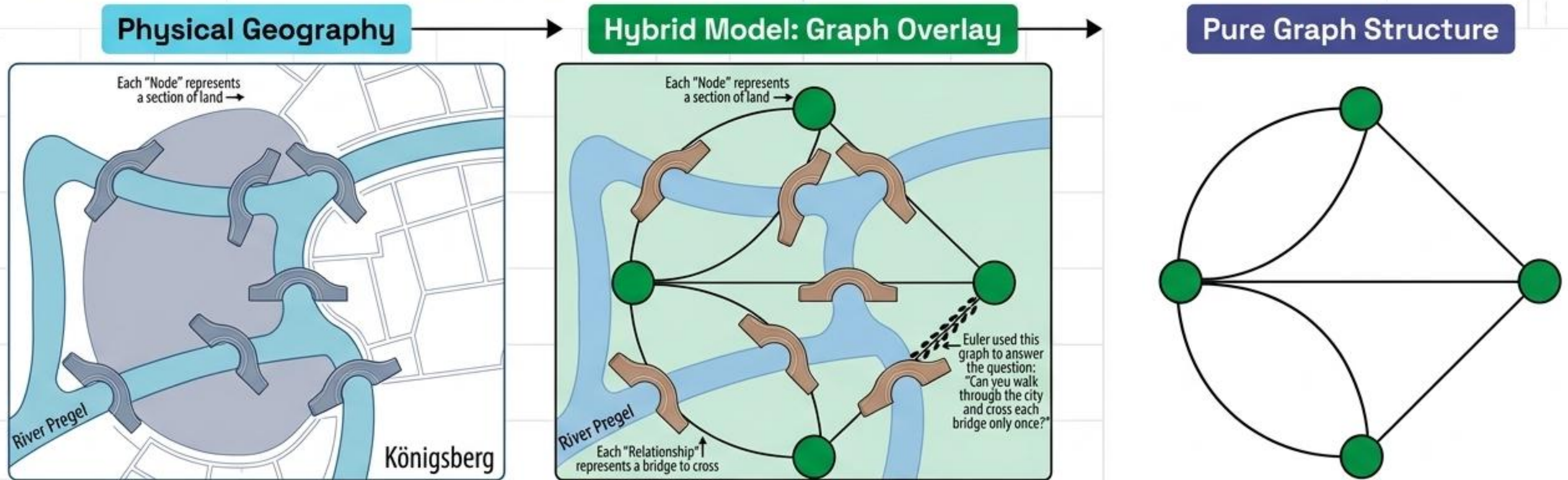


Mapping how discrete entities connect and relate to one another.

In this context, graphs (sometimes referred to as networks) are not about visualizing numbers—they are a structural way to map connections.

The Origins of Abstraction: Stripping Away the Noise

In the 18th century, Leonhard Euler solved the Königsberg bridge problem not by walking, but by reducing physical geography to a logical form.



Like modern software development, graphs strip away the inherent noise of the real world, concentrating solely on how things are connected.

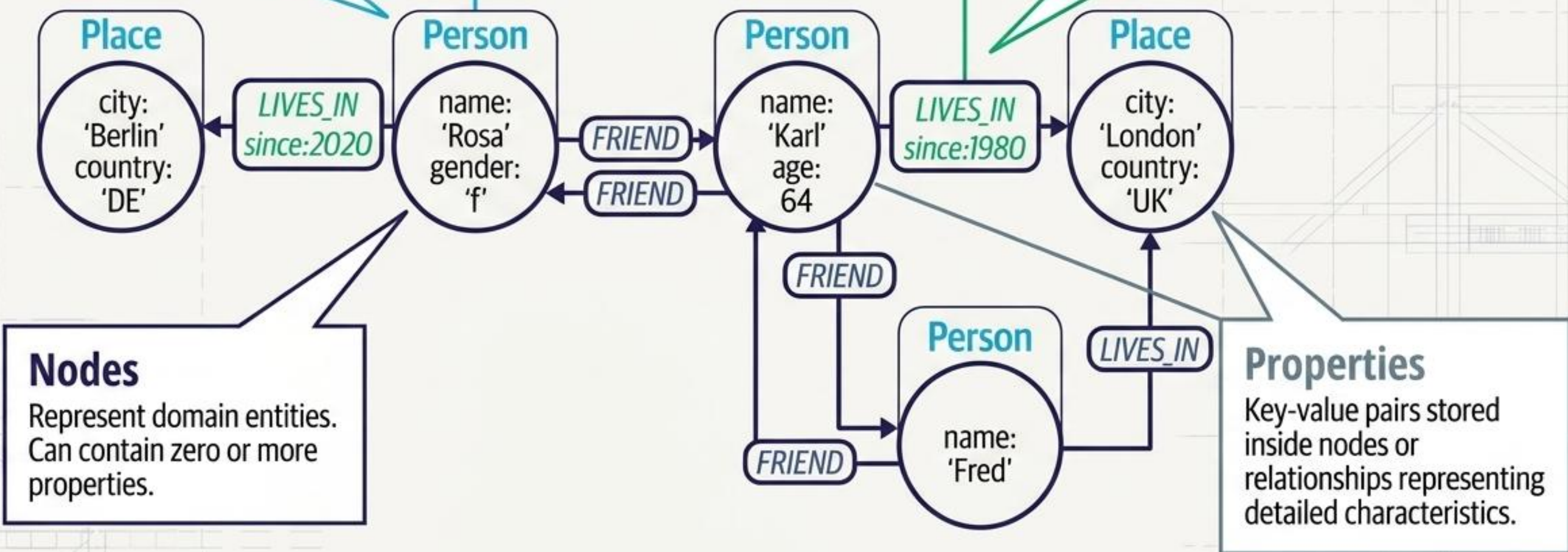
The Anatomy of a Labeled Property Graph

Labels

Declare the node's purpose or role in the overarching graph.

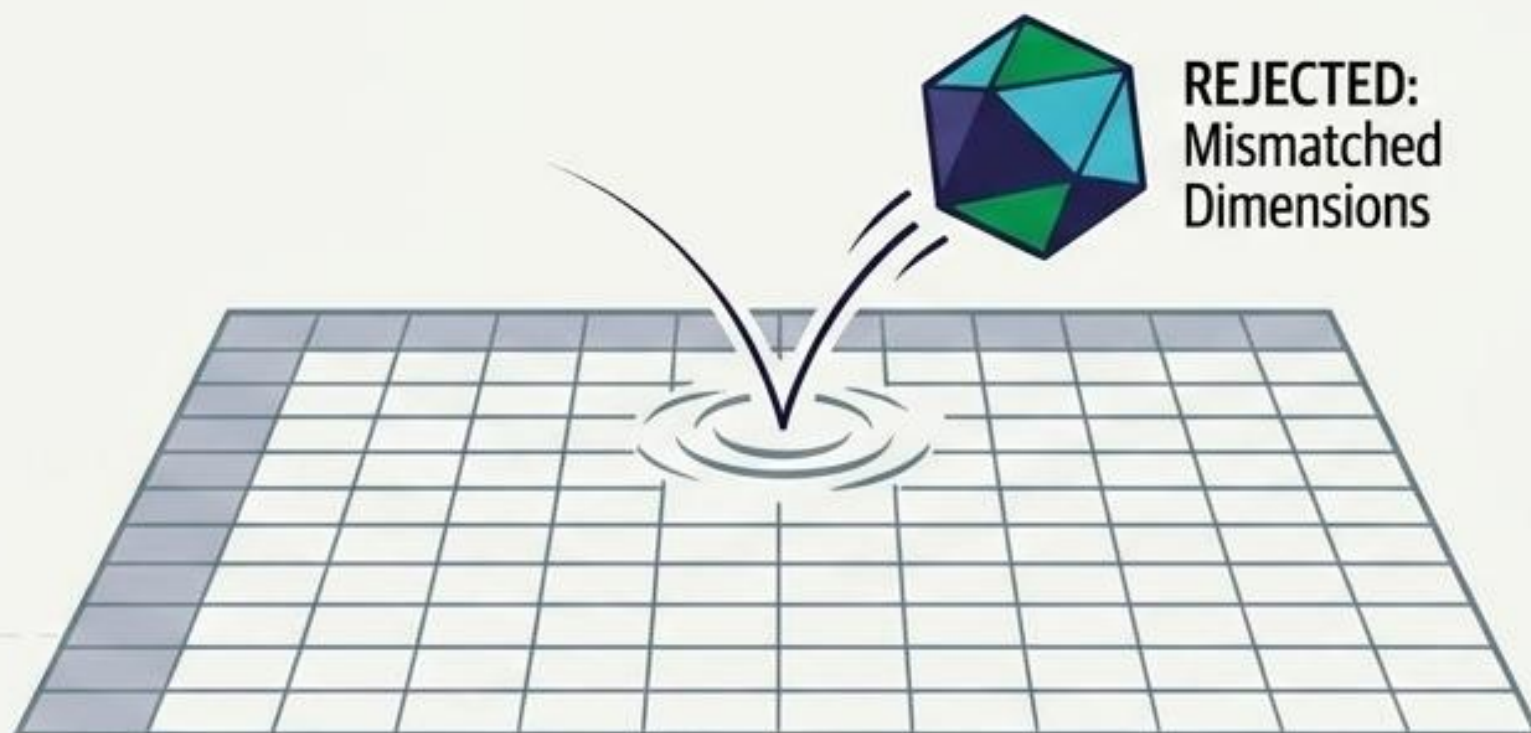
Relationships

Show how entities interrelate. They have a specific type, a strict direction, and never dangle.



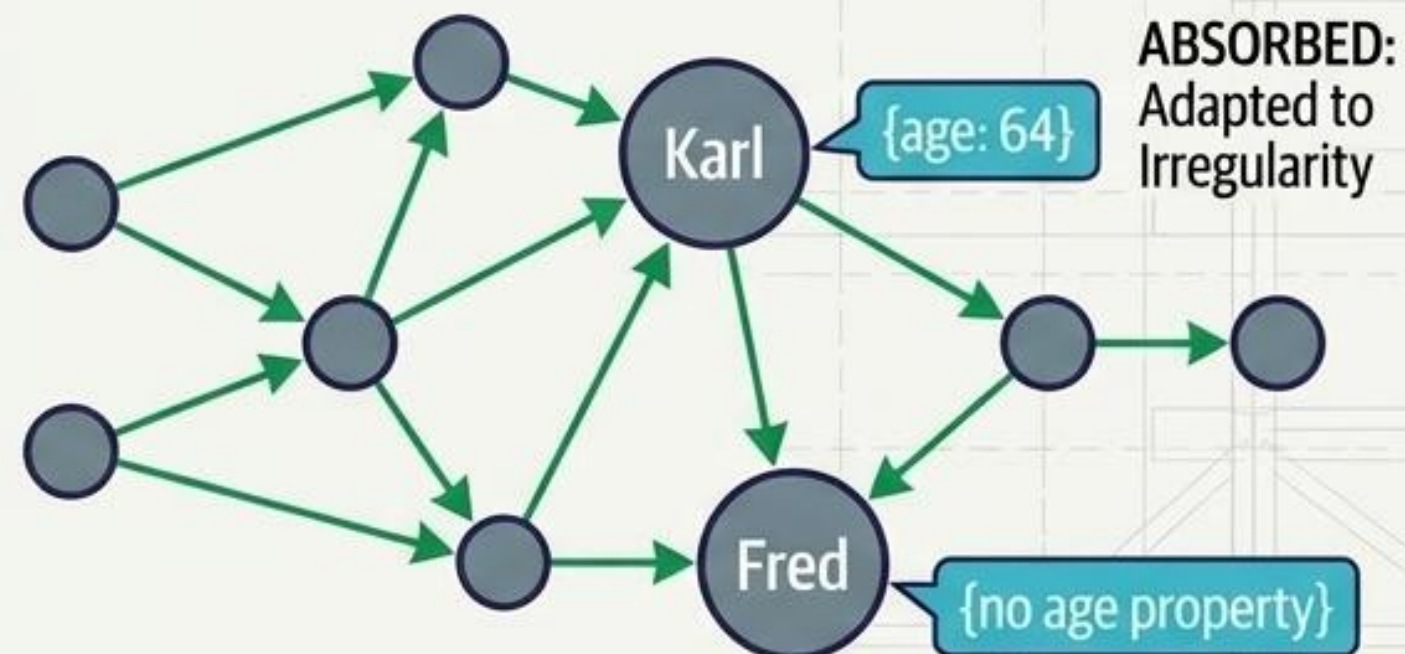
The Power of Schema-ish Flexibility

Rigid Schema (e.g., RDBMS)



Inflexible structures cannot accommodate unique or incomplete data.

Property Graph Network (Schema-ish)



Flexible models easily incorporate diverse data structures.

Real-world data is inherently uneven and incomplete.

Property graphs do not enforce rigid schemas across the entire model. Constraints are applied only where necessary.

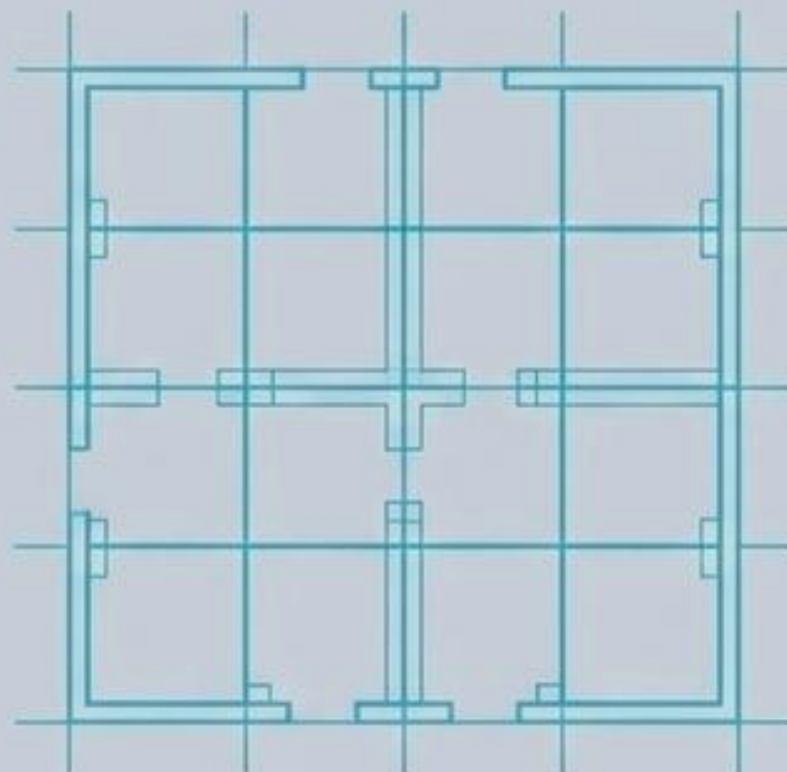
This 'schema-ish' approach easily accommodates the messiness of reality, allowing the model to evolve continuously.

What Makes a Graph a Knowledge Graph?

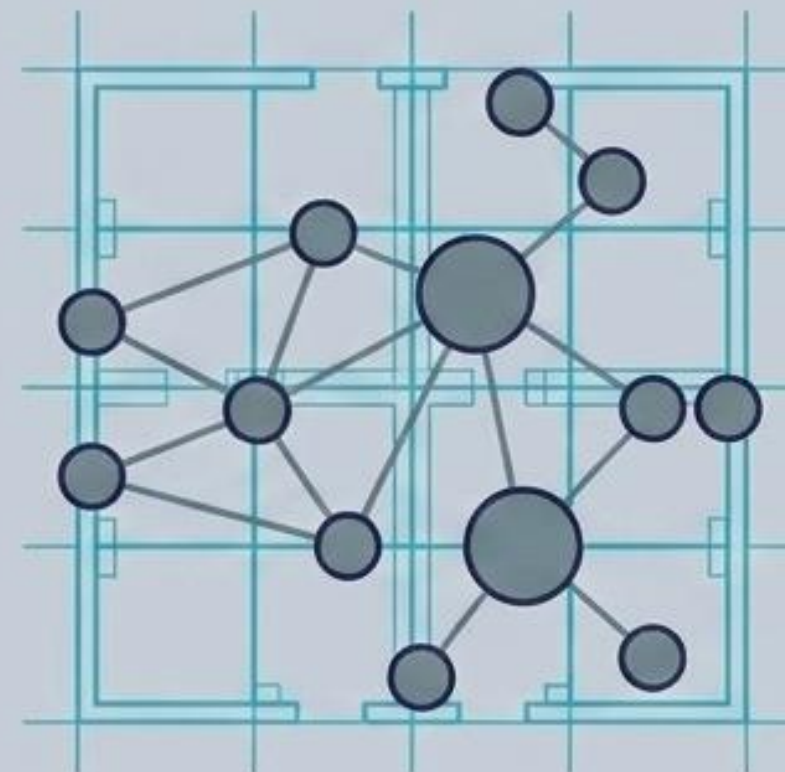
[The Property Graph]



[The Organizing Principle]

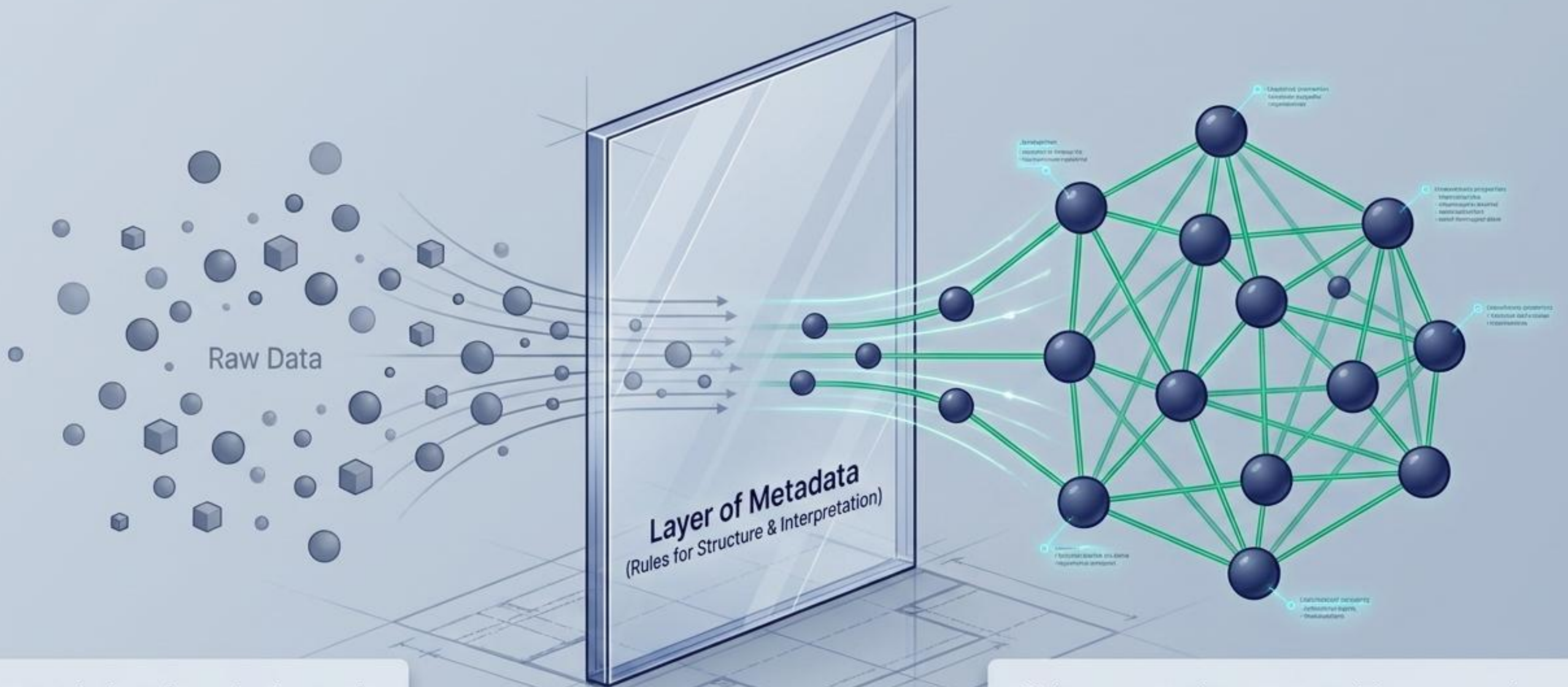


[The Knowledge Graph]



A Knowledge Graph is an interlinked set of facts enriched with an organizing principle. This added layer of structure supports human and machine reasoning, upgrading raw data into discoverable knowledge.

The Context Lens: Deriving Meaning Through Metadata



"Context derives from the layer of metadata that provides rules for structure and interpretation."

This connected context enables responsive businesses to drive automation, process optimization, and improve predictive accuracy.

The Intelligence Inversion

Where does the intelligence live?

The Traditional Paradigm

Data

Dumb

(Raw material waiting to be mined;
rigidly formatted)

Applications

Highly Complex

(Must contain all the intelligence and
business logic to interpret the data)

The Knowledge Graph Paradigm

Smart

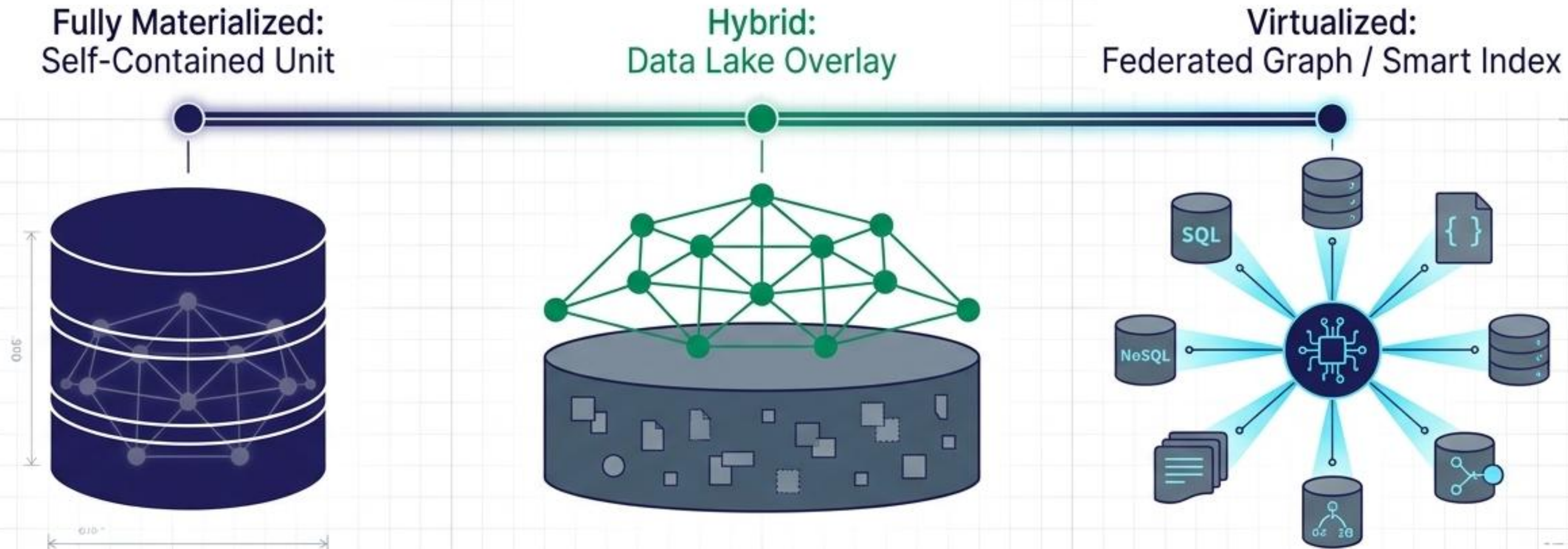
(Self-describing, enriched with an
organizing principle and context)

Simpler & Reusable

(Because the data inherently explains
how it connects, downstream
systems become vastly simpler)

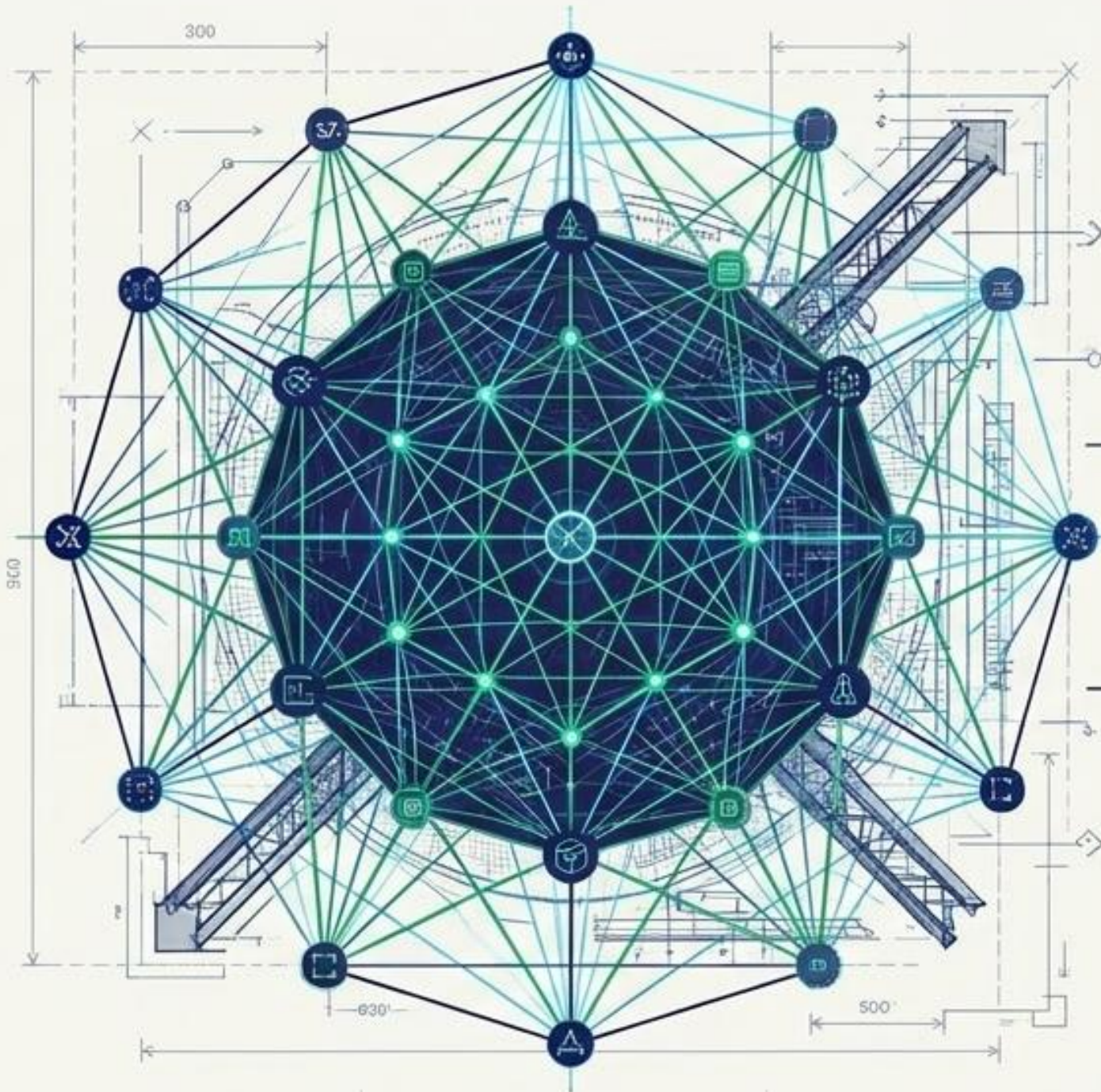
Having smarter data both simplifies systems and encourages broad reuse across the enterprise.

Architectural Agnosticism: Where Does the Data Live?



Knowledge graphs are agnostic about physical storage. They scale from smart indexes over external data to fully hosted graph platforms.

The Blueprint Complete: Your Data's Curator



At its core, a Knowledge Graph acts as a rich index over your enterprise data.

It provides curation, context, and connection—much like a skilled librarian effortlessly recommending the exact journals and insights needed to solve a complex problem.

Moving from **scattered data** to an **interconnected organizing principle** is the foundation for the agile, responsive business.